

HealthVault

API Documentation

Backend Integration Guide for Frontend Teams

Version	v1.0
Base URL	http://127.0.0.1:8000/api/v1/
Auth	Bearer JWT Token
Date	May 2026

Admin Portal

Lab / Hospital Portal

User / Patient Portal

Health records,
management

Overview

This document describes all REST API endpoints for the HealthVault platform. It is intended for frontend developers integrating the Admin, Lab/Hospital, and Patient portals.

Authentication: All endpoints except Auth routes require a Bearer JWT token in the Authorization header. Tokens are portal-specific — an admin token cannot be used on lab endpoints and vice versa.

Base URL: `http://127.0.0.1:8000/api/v1/` (development)

Content-Type: `application/json` for all requests except file upload which uses `multipart/form-data`.

Trailing slashes: All URLs require a trailing slash.

Method Color Reference

GET	Read / Retrieve data
POST	Create or trigger an action
PATCH	Partial update of a resource
DELETE	Remove a resource

Standard Pagination Response

```
{ "count": 42, "total_pages": 3, "current_page": 1, "next": "http://.../api/v1/.../?page=2",  
  "previous": null, "results": [ ... ] }
```

Field Value Reference

Field	Allowed Values
blood_group	A+ A- B+ B- AB+ AB- O+ O-
gender	M (Male) F (Female) O (Other) N (Prefer not to say)
type	LAB HOSPITAL
channels	INAPP EMAIL SMS WHATSAPP
consent status	PENDING OTP_VERIFIED ACTIVE EXPIRED CANCELLED
datetime format	2026-01-15T10:30:00 (ISO 8601)
date format	2026-01-15 (YYYY-MM-DD)
phone format	+919876543210 (E.164 with country code)

Authentication Endpoints

These endpoints are open — no Authorization header required. All other endpoints require a Bearer token obtained from these endpoints.

POST /api/v1/auth/admin/login/

OPEN

Admin login with username and password.

Request Body

```
{ "username": "admin", "password": "yourpassword" }
```

Sample Response

```
{ "access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "refresh":  
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "portal": "admin" }
```

POST /api/v1/auth/lab/login/

OPEN

Lab or Hospital login with phone and password.

Request Body

```
{ "phone": "+918087209688", "password": "labpassword123" }
```

Sample Response

```
{ "access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "refresh":  
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "portal": "lab" }
```

POST /api/v1/auth/user/register/

OPEN

Patient self-registration. Only name and phone are required.

Request Body

```
{ "name": "Rahul Sharma", "phone": "+919876543210", "email": "rahul@example.com",  
"birth_date": "1995-06-15", "blood_group": "B+", "gender": "M", "address": "Karvenagar,  
Pune" }
```

Sample Response

```
{ "detail": "Account created. Please login using OTP.", "user_uid": "929765985864" }
```

POST

[/api/v1/auth/user/request-otp/](#)

OPEN

Send OTP to patient's WhatsApp and SMS. Rate limited — 1 request per 60 seconds.

Request Body

```
{ "phone": "+919876543210" }
```

Sample Response

```
{ "detail": "OTP sent to your WhatsApp and SMS." }
```

Note: OTP expires in 5 minutes. Maximum 5 verification attempts per OTP.

POST

[/api/v1/auth/user/verify-otp/](#)

OPEN

Verify OTP and receive JWT token. After 5 failed attempts, user must request a new OTP.

Request Body

```
{ "phone": "+919876543210", "otp": "482910" }
```

Sample Response

```
{ "access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "refresh":  
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "portal": "user" }
```

POST

[/api/v1/auth/token/refresh/](#)

OPEN

Get a new access token using the refresh token.

Request Body

```
{ "refresh": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." }
```

Sample Response

```
{ "access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." }
```

Admin Portal Endpoints

All endpoints require: Authorization: Bearer <admin_token>

Labs & Hospitals

GET

/api/v1/admin/labs/

ADMIN

List all registered labs and hospitals with search, sort, and filter.

Query Parameters

Field	Type	Required	Description
search	string	No	Search by name, email, phone, lab_id
type	string	No	Filter by LAB or HOSPITAL
is_active	boolean	No	Filter by active status: true or false
ordering	string	No	Sort by: name, -name, created_at, -created_at, plan_end, storage_used_mb
page	integer	No	Page number (default: 1)

Sample Response

```
{ "count": 2, "total_pages": 1, "current_page": 1, "next": null, "previous": null,
  "results": [ { "id": "uuid", "lab_id": "359878627832", "name": "City Pathlab", "email":
    "city@pathlab.com", "phone": "+918087209688", "type": "LAB", "is_active": true,
    "is_plan_active": true, "storage_used_gb": 0.0006, "storage_limit_gb": 50.0, "plan_end":
    "2027-05-16T01:00:00+05:30", "report_count": 2 } ] }
```

POST

/api/v1/admin/labs/

ADMIN

Register a new lab or hospital. subscription_plan_id is optional.

Request Body

```
{ "name": "City Pathlab", "email": "city@pathlab.com", "phone": "+918087209688",
  "password": "labpassword123", "address": "Karvenagar, Pune", "type": "LAB",
  "plan_duration_days": 365, "storage_gb": 50, "subscription_plan_id": "uuid-or-null" }
```

Sample Response

```
{ "id": "uuid", "lab_id": "359878627832", "name": "City Pathlab", "type": "LAB",
  "is_active": true, "storage_limit_gb": 50.0, "plan_start": "2026-05-16T01:00:00+05:30",
  "plan_end": "2027-05-16T01:00:00+05:30" }
```

GET

`/api/v1/admin/labs/:id/`

ADMIN

Full detail of a specific lab including storage stats and report count.

Sample Response

```
{ "id": "uuid", "lab_id": "359878627832", "name": "City Pathlab", "email":  
"city@pathlab.com", "phone": "+918087209688", "address": "Karvenagar, Pune", "type":  
"LAB", "is_active": true, "is_plan_active": true, "storage_used_mb": 0.6,  
"storage_used_gb": 0.0006, "storage_limit_mb": 51200.0, "storage_limit_gb": 50.0,  
"plan_start": "2026-05-16T01:00:00+05:30", "plan_end": "2027-05-16T01:00:00+05:30",  
"subscription_plan_name": "Standard Plan", "report_count": 2, "created_at":  
"2026-05-16T01:00:00+05:30" }
```

PATCH

`/api/v1/admin/labs/:id/`

ADMIN

Update lab info, storage limit, plan dates, or active status. Send only fields to update.

Request Body

```
{ "storage_gb": 100, "plan_end": "2028-01-01T00:00:00", "is_active": true }
```

Sample Response

```
{ "id": "uuid", "name": "City Pathlab", "storage_limit_gb": 100.0, "plan_end":  
"2028-01-01T00:00:00+05:30", "is_active": true }
```

POST

`/api/v1/admin/labs/:id/toggle-active/`

ADMIN

Toggle lab active status — activates if inactive, deactivates if active. No body required.

Sample Response

```
{ "detail": "Lab deactivated successfully.", "is_active": false }
```

Users / Patients

GET

[/api/v1/admin/users/](#)

ADMIN

List all registered patients.

Query Parameters

Field	Type	Required	Description
search	string	No	Search by name, phone, email, user_uid
ordering	string	No	Sort by: name, -name, created_at, -created_at
page	integer	No	Page number

Sample Response

```
{ "count": 1, "results": [ { "id": "uuid", "user_uid": "929765985864", "name": "Rahul Sharma", "phone": "+919876543210", "email": "rahul@example.com", "blood_group": "B+", "gender": "M", "is_active": true, "created_at": "2026-05-16T01:00:00+05:30" } ] }
```

POST

[/api/v1/admin/users/:id/toggle-active/](#)

ADMIN

Toggle patient account active status. No body required.

Sample Response

```
{ "detail": "User deactivated successfully.", "is_active": false }
```

Subscription Plans

GET

[/api/v1/subscriptions/](#)

OPEN

List all subscription plan templates. Open to all portals.

Sample Response

```
{ "results": [ { "id": "uuid", "name": "Standard Plan", "storage_gb": 50, "duration_days": 365, "description": "Standard yearly plan" } ] }
```

POST

[/api/v1/subscriptions/](#)

ADMIN

Create a new subscription plan template.

Request Body

```
{ "name": "Standard Plan", "storage_gb": 50, "duration_days": 365, "description": "Standard yearly plan" }
```

Sample Response

```
{ "id": "uuid", "name": "Standard Plan", "storage_gb": 50, "duration_days": 365, "description": "Standard yearly plan", "created_at": "2026-05-16T01:00:00+05:30" }
```

PATCH

[/api/v1/subscriptions/:id/](#)

ADMIN

Update a subscription plan template.

Request Body

```
{ "storage_gb": 100, "duration_days": 180 }
```

Sample Response

```
{ "id": "uuid", "name": "Standard Plan", "storage_gb": 100, "duration_days": 180 }
```

DELETE

[/api/v1/subscriptions/:id/](#)

ADMIN

Delete a subscription plan template.

Sample Response

```
HTTP 204 No Content
```

Notifications

POST

[/api/v1/notifications/send/](#)

ADMIN

Send a message to a lab via one or more channels simultaneously.

Request Body

```
{ "lab_id": "uuid", "message": "Your subscription expires in 30 days. Please renew.",  
  "channels": ["INAPP", "EMAIL", "WHATSAPP"] }
```

Sample Response

```
[ {"id": "uuid", "channel": "INAPP", "status": "SENT"}, {"id": "uuid", "channel": "EMAIL",  
  "status": "SENT"}, {"id": "uuid", "channel": "WHATSAPP", "status": "SENT"} ]
```

Note: Channels: INAPP, EMAIL, SMS, WHATSAPP. Pick one or more.

GET

</api/v1/notifications/>

ADMIN

List all notifications sent. Filter by lab.

Query Parameters

Field	Type	Required	Description
lab_id	uuid	No	Filter notifications for a specific lab

Sample Response

```
{ "results": [ { "id": "uuid", "lab_name": "City Pathlab", "message": "Your subscription expires in 30 days.", "channel": "INAPP", "status": "SENT", "created_at": "2026-05-16T01:00:00+05:30" } ] }
```

Analytics

GET

</api/v1/analytics/dashboard/>

ADMIN

Platform-wide aggregate statistics for the admin dashboard.

Sample Response

```
{ "labs": { "total": 5, "active": 4, "inactive": 1, "plan_expired": 0 }, "users": { "total": 120, "active": 118 }, "storage": { "total_used_mb": 1240.5, "total_used_gb": 1.211, "total_limit_gb": 250.0, "utilisation_percent": 0.5 }, "reports": { "total": 340, "total_size_mb": 1240.5 }, "consents": { "total": 28, "active": 5, "expired": 23 } }
```

GET

</api/v1/analytics/storage/>

ADMIN

Per-lab storage breakdown sorted by usage — useful for billing decisions.

Sample Response

```
{ "count": 2, "results": [ { "lab_id": "359878627832", "name": "City Pathlab", "type": "LAB", "storage_used_gb": 0.8, "storage_limit_gb": 50.0, "utilisation_percent": 1.6, "report_count": 45, "plan_end": "2027-05-16T01:00:00+05:30", "is_active": true } ] }
```

Lab / Hospital Portal Endpoints

All endpoints require: Authorization: Bearer <lab_token>

Dashboard / Profile

GET /api/v1/lab/dashboard/

LAB

Lab self-profile with storage stats and report/user counts.

Sample Response

```
{ "id": "uuid", "lab_id": "359878627832", "name": "City Pathlab", "email": "city@pathlab.com", "phone": "+918087209688", "type": "LAB", "is_active": true, "is_plan_active": true, "plan_start": "2026-05-16T01:00:00+05:30", "plan_end": "2027-05-16T01:00:00+05:30", "subscription_plan": "Standard Plan", "storage": { "used_mb": 0.6, "used_gb": 0.0006, "limit_gb": 50.0, "utilisation_percent": 0.0 }, "stats": { "total_reports": 2, "total_users": 1 } }
```

Reports

GET /api/v1/lab/reports/

LAB

List all reports — own uploaded and consent-based. Filter by type.

Query Parameters

Field	Type	Required	Description
type	string	No	own consented expired_consent all (default: all)
search	string	No	Search by filename, description, patient name/phone
ordering	string	No	Sort by: created_at, issued_date, file_size_mb
page	integer	No	Page number

Sample Response

```
{ "count": 2, "results": [ { "id": "uuid", "report_uid": "092429554873", "original_filename": "blood_test.pdf", "description": "Blood Test Report", "file_size_mb": 0.32, "issued_date": "2026-05-15", "created_at": "2026-05-16T02:40:30+05:30", "user_name": "Rahul Sharma", "user_uid": "929765985864", "user_phone": "+919876543210", "is_consent_based": false, "consent_expires_at": null, "is_deleted": false } ] }
```

POST

`/api/v1/lab/reports/upload/`

LAB

Upload a PDF report for a patient. Use multipart/form-data. File is uploaded to cloud storage automatically.

Request Body

```
Content-Type: multipart/form-data file → [select PDF file] (required, max 50MB) user_phone → +919876543210 (required) description → Blood Test Report (optional) issued_date → 2026-05-15 (optional, YYYY-MM-DD)
```

Sample Response

```
{ "id": "uuid", "report_uid": "092429554873", "original_filename": "blood_test.pdf", "file_size_mb": 0.32, "issued_date": "2026-05-15", "user_name": "Rahul Sharma", "user_uid": "929765985864", "created_at": "2026-05-16T02:40:30+05:30" }
```

Note: Do NOT set Content-Type header manually — let the HTTP client set it with the boundary automatically.

GET

`/api/v1/lab/reports/:id/`

LAB

Full detail of a specific report.

Sample Response

```
{ "id": "uuid", "report_uid": "092429554873", "original_filename": "blood_test.pdf", "description": "Blood Test Report", "file_size_mb": 0.32, "issued_date": "2026-05-15", "user_name": "Rahul Sharma", "user_phone": "+919876543210", "is_consent_based": false, "is_deleted": false }
```

GET

`/api/v1/lab/reports/:id/stream/`

LAB

Get a short-lived signed URL (60 seconds) to view the PDF inline in the browser. Open this URL in an iframe or new tab.

Sample Response

```
{ "url": "https://minio.example.com/reports/...?X-Amz-Expires=60&...", "expires_in_seconds": 60, "filename": "blood_test.pdf" }
```

Note: URL expires in 60 seconds. Generate a fresh URL each time the user wants to view. Do not cache or store this URL.

Patients

GET

/api/v1/lab/users/

LAB

List all patients who have reports in this lab.

Query Parameters

Field	Type	Required	Description
search	string	No	Search by name, phone, user_uid
ordering	string	No	Sort by: name, created_at

Sample Response

```
{ "results": [ { "id": "uuid", "user_uid": "929765985864", "name": "Rahul Sharma",  
"phone": "+919876543210", "blood_group": "B+", "gender": "M", "is_active": true } ] }
```

GET

/api/v1/lab/users/:id/reports/

LAB

All reports for a specific patient — own uploaded, active consented, and expired consent metadata.

Sample Response

```
{ "results": [ { "id": "uuid", "report_uid": "092429554873", "original_filename":  
"blood_test.pdf", "is_consent_based": true, "consent_expires_at":  
"2026-12-31T23:59:59+05:30" } ] }
```

Consent Management

POST

/api/v1/lab/consent/request/

LAB

Initiate a consent request to a patient. OTP is sent to patient's WhatsApp automatically. OTP and consent window expire in 5 minutes.

Request Body

```
{ "user_phone": "+919876543210", "description": "Requesting access to previous blood test  
reports for treatment" }
```

Sample Response

```
{ "id": "uuid", "to_user": "Rahul Sharma", "description": "Requesting access to previous  
blood test reports", "status": "PENDING", "otp_expires_at": "2026-05-16T03:05:00+05:30",  
"created_at": "2026-05-16T03:00:00+05:30" }
```

GET

[/api/v1/lab/consent/](#)

LAB

List all consent requests sent by this lab.

Query Parameters

Field	Type	Required	Description
status	string	No	PENDING OTP_VERIFIED ACTIVE EXPIRED CANCELLED
search	string	No	Search by patient name, phone, user_uid

Sample Response

```
{ "results": [ { "id": "uuid", "user_name": "Rahul Sharma", "user_uid": "929765985864",  
"description": "Requesting previous reports", "status": "ACTIVE", "is_access_live": true,  
"consent_expires_at": "2026-12-31T23:59:59+05:30", "file_count": 2 } ] }
```

GET

[/api/v1/lab/consent/:id/](#)

LAB

Consent request detail including list of shared files.

Sample Response

```
{ "id": "uuid", "user_name": "Rahul Sharma", "status": "ACTIVE", "is_access_live": true,  
"consent_expires_at": "2026-12-31T23:59:59+05:30", "files": [ { "report_uid":  
"041210841147", "filename": "blood_test.pdf", "issued_date": "2026-05-15", "expires_at":  
"2026-12-31T23:59:59+05:30", "is_active": true } ] }
```

GET

[/api/v1/lab/notifications/](#)

LAB

In-app notifications sent by admin to this lab.

Sample Response

```
{ "results": [ { "id": "uuid", "message": "Your subscription expires in 30 days. Please  
renew.", "channel": "INAPP", "status": "SENT", "created_at": "2026-05-16T01:00:00+05:30" }  
] }
```

Patient / User Portal Endpoints

All endpoints require: Authorization: Bearer <user_token>

Profile

GET /api/v1/user/profile/

USER

Retrieve own profile details.

Sample Response

```
{ "id": "uuid", "user_uid": "929765985864", "name": "Rahul Sharma", "email": "rahul@example.com", "phone": "+919876543210", "birth_date": "1995-06-15", "blood_group": "B+", "gender": "M", "address": "Karvenagar, Pune", "is_active": true, "created_at": "2026-05-16T01:00:00+05:30" }
```

PATCH /api/v1/user/profile/

USER

Update own profile. Send only the fields you want to change.

Request Body

```
{ "name": "Rahul Sharma", "email": "rahul.new@example.com", "birth_date": "1995-06-15", "blood_group": "O+", "gender": "M", "address": "Kothrud, Pune" }
```

Sample Response

```
{ "name": "Rahul Sharma", "email": "rahul.new@example.com", "blood_group": "O+", "address": "Kothrud, Pune" }
```

Reports / Health Records

GET

`/api/v1/user/reports/`

USER

List all reports uploaded for this patient across all labs.

Query Parameters

Field	Type	Required	Description
search	string	No	Search by filename, description, lab name
ordering	string	No	Sort by: created_at, issued_date, original_filename
page	integer	No	Page number

Sample Response

```
{ "count": 2, "results": [ { "id": "uuid", "report_uid": "092429554873",  
  "original_filename": "blood_test.pdf", "description": "Blood Test Report", "file_size_mb":  
  0.32, "issued_date": "2026-05-15", "created_at": "2026-05-16T02:40:30+05:30", "lab_name":  
  "City Pathlab", "lab_id": "359878627832" } ] }
```

GET

`/api/v1/user/reports/:id/stream/`

USER

Get a short-lived signed URL (60 seconds) to view the report PDF inline in the browser.

Sample Response

```
{ "url": "https://minio.example.com/reports/...?X-Amz-Expires=60&...",  
  "expires_in_seconds": 60, "filename": "blood_test.pdf" }
```

Note: URL expires in 60 seconds. Open in an iframe or new browser tab. Generate a fresh URL each time.

GET

`/api/v1/user/reports/:id/download/`

USER

Get a download URL (300 seconds TTL) for the report PDF with attachment disposition.

Sample Response

```
{ "url": "https://minio.example.com/reports/...?X-Amz-Expires=300&...",  
  "expires_in_seconds": 300, "filename": "blood_test.pdf" }
```

Consent Management

The consent flow has 3 steps: (1) User sees incoming request, (2) User verifies OTP received on WhatsApp, (3) User selects files and sets expiry.

GET

/api/v1/user/consent/

USER

List incoming consent requests. Default shows only live/pending. Use ?history=true for full history.

Query Parameters

Field	Type	Required	Description
history	boolean	No	true = include EXPIRED and CANCELLED requests

Sample Response

```
{ "results": [ { "id": "uuid", "lab_name": "City Pathlab", "lab_id": "359878627832",
"description": "Requesting previous blood test reports", "status": "PENDING",
"is_access_live": false, "otp_expires_at": "2026-05-16T03:05:00+05:30", "created_at":
"2026-05-16T03:00:00+05:30" } ] }
```

POST

/api/v1/user/consent/:id/verify-otp/

USER

Step 1: Verify the OTP received on WhatsApp. Maximum 5 attempts. OTP window is 5 minutes.

Request Body

```
{ "otp": "482910" }
```

Sample Response

```
{ "detail": "OTP verified. Please select files to share.", "consent_id": "uuid" }
```

Note: After success, consent status moves to OTP_VERIFIED. Then call /share/ to complete.

POST

/api/v1/user/consent/:id/share/

USER

Step 2: Select which reports to share and set the expiry date. Consent must be in OTP_VERIFIED status.

Request Body

```
{ "report_ids": [ "uuid-of-report-1", "uuid-of-report-2" ], "expires_at":
"2026-12-31T23:59:59" }
```

Sample Response

```
{ "id": "uuid", "status": "ACTIVE", "consent_expires_at": "2026-12-31T23:59:59+05:30",
"files": [ { "filename": "blood_test.pdf", "expires_at": "2026-12-31T23:59:59+05:30" } ] }
```

Note: expires_at must be a future datetime. Lab loses access automatically after this time.

POST**/api/v1/user/consent/:id/revoke/**

USER

Immediately revoke an active consent. Lab loses access instantly. No body required.

Sample Response

```
{ "detail": "Consent revoked successfully." }
```

GET**/api/v1/user/consent/active-shares/**

USER

See which labs currently have live access to which of your files.

Sample Response

```
{ "results": [ { "id": "uuid", "lab_name": "City Pathlab", "status": "ACTIVE",  
  "is_access_live": true, "consent_expires_at": "2026-12-31T23:59:59+05:30", "files": [ {  
    "filename": "blood_test.pdf", "issued_date": "2026-05-15", "expires_at":  
    "2026-12-31T23:59:59+05:30" } ] } ] }
```

Error Reference

HTTP Code	Meaning	Common Cause
200 OK	Success	Request processed successfully
201 Created	Resource created	POST request created a new record
204 No Content	Success, no body	DELETE request completed
400 Bad Request	Validation error	Missing required field, invalid value, OTP wrong
401 Unauthorized	Auth failed	Missing or invalid Bearer token
403 Forbidden	Wrong portal	Using admin token on lab endpoint or vice versa
404 Not Found	Resource missing	Invalid ID or record does not belong to you
429 Too Many Requests	Rate limited	OTP max attempts (5) or resend cooldown (60s)
500 Internal Server Error	Server error	Contact backend team with request details

Sample Error Response

```
{ "detail": "OTP expired or not found. 3 attempts remaining." } // Validation error { "phone": ["No active account found for this number."], "otp": ["This field is required."] }
```